

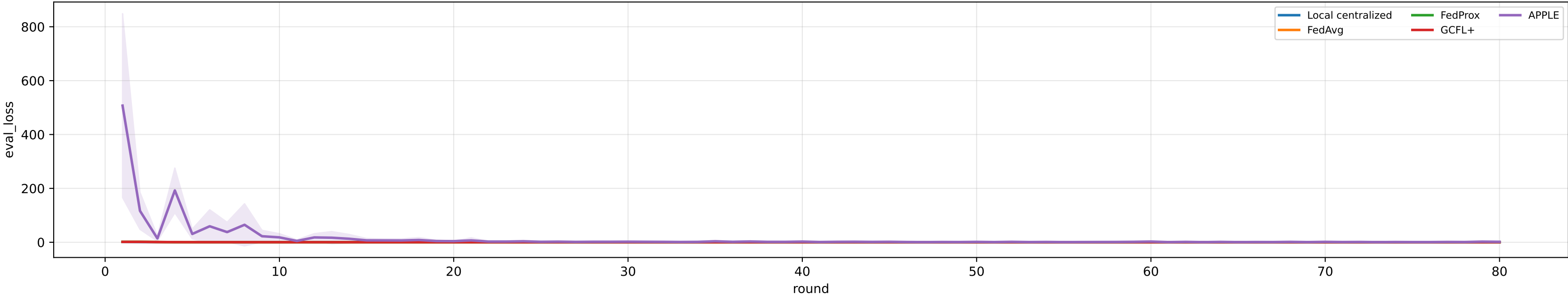
Controlled task-specialization level: High | aggregated / pooled performance over 5 clients  
| model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64

family=specialized\_cycle2 | n\_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |  
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=specialized\_cycle3 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |  
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=specialized\_cycle4 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% |  
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=specialized\_cycle5 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |  
cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=specialized\_cycle6 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |  
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3%  
specialization=high | specialization\_fraction=0.80 | global\_visible\_support=0.96 | task\_jsd\_mean=0.162 | task\_jsd\_median=0.162 | task\_jsd\_max=0.165 |  
family\_centroid\_jsd\_mean=0.162

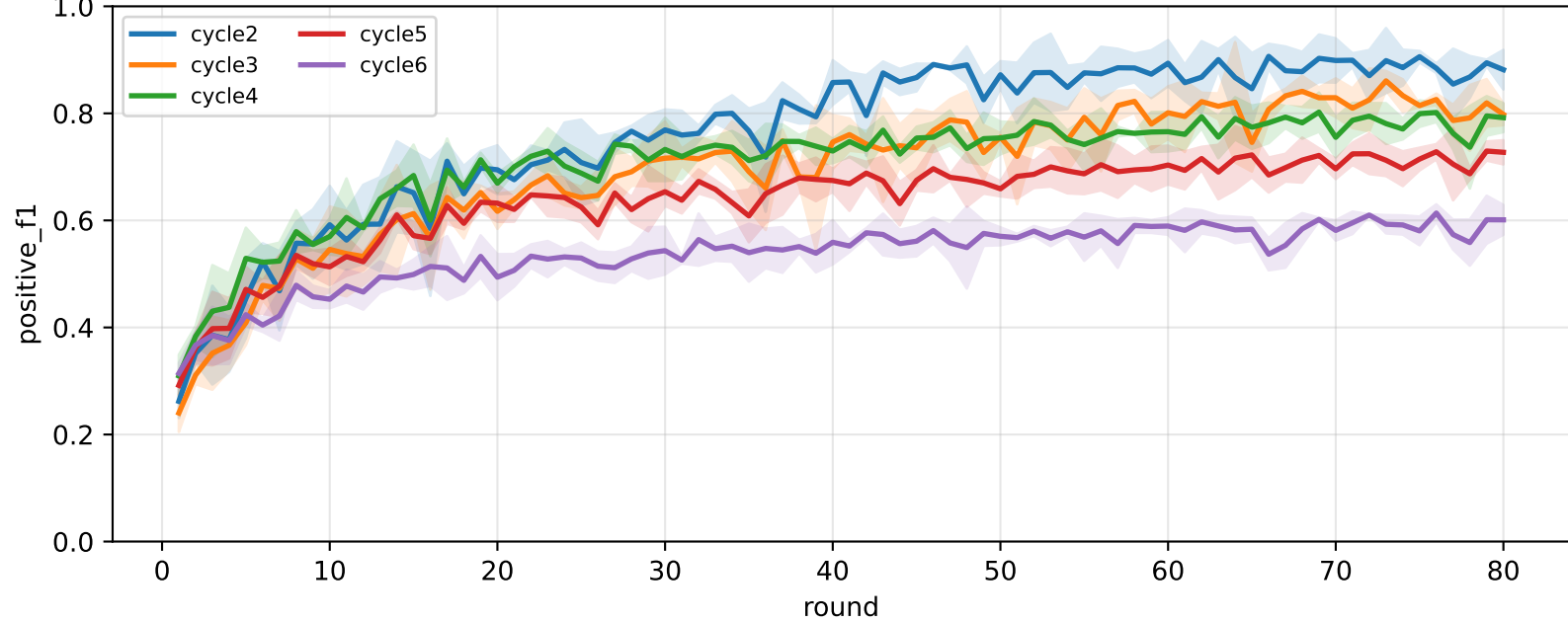
Best logged test performance (mean ± std over seeds)

| Algorithm         | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Local centralized | 74.7% ± 1.4%        | 78.7% ± 1.0%        | 91.5% ± 2.6%        | 89.1% ± 0.9%        | 80.9% ± 1.9%        | 68.6% ± 3.9%        | 58.8% ± 1.7%        |
| FedAvg            | 76.8% ± 1.8%        | <b>80.5% ± 1.9%</b> | 95.1% ± 1.3%        | 92.3% ± 3.1%        | 81.6% ± 1.4%        | <b>74.6% ± 3.1%</b> | 59.0% ± 1.0%        |
| FedProx           | 76.0% ± 1.4%        | 80.0% ± 1.0%        | 94.9% ± 0.7%        | <b>92.6% ± 2.2%</b> | <b>83.5% ± 1.6%</b> | 70.0% ± 3.6%        | 58.9% ± 2.2%        |
| GCFL+             | <b>77.8% ± 1.7%</b> | <b>80.5% ± 1.3%</b> | <b>95.5% ± 0.6%</b> | 90.2% ± 0.9%        | 78.7% ± 2.0%        | 72.8% ± 0.8%        | <b>65.0% ± 3.6%</b> |
| APPLE             | 57.2% ± 4.1%        | 61.6% ± 2.7%        | 75.1% ± 6.2%        | 69.1% ± 9.7%        | 59.1% ± 6.2%        | 57.7% ± 8.7%        | 43.7% ± 5.3%        |

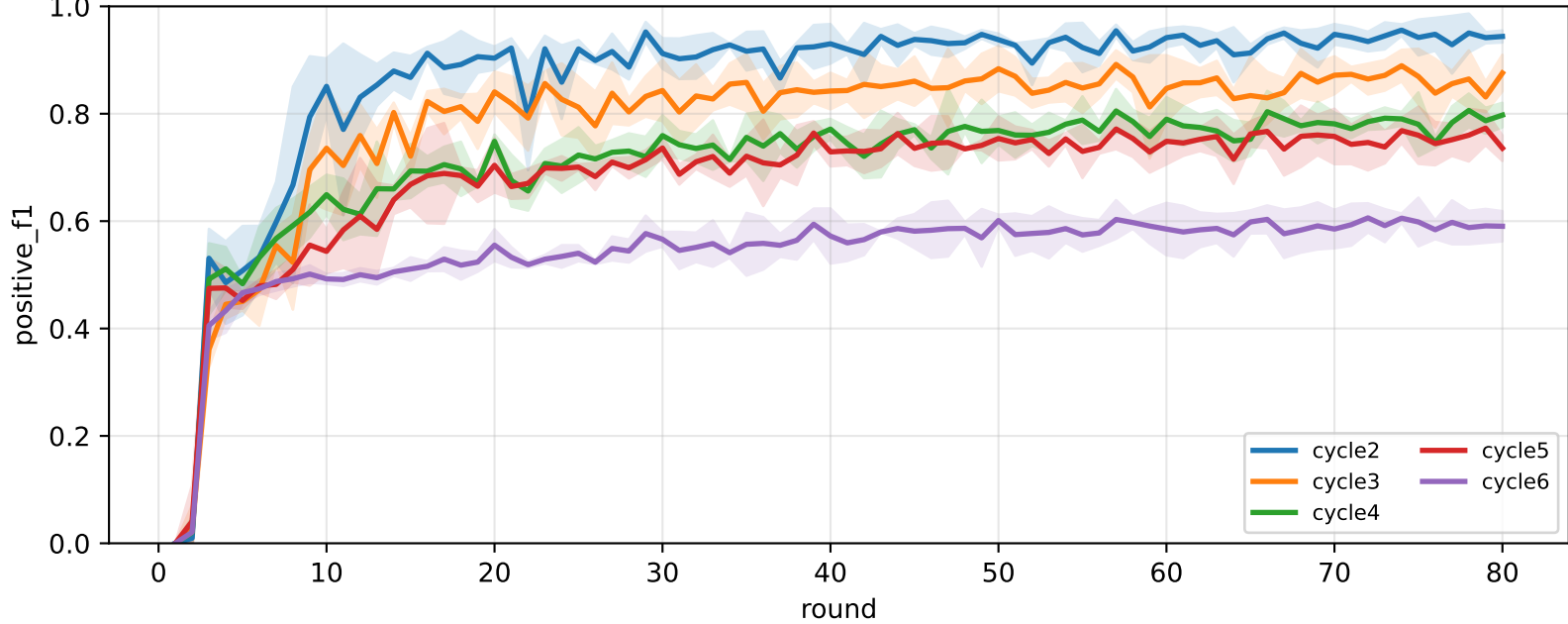
Validation loss comparison



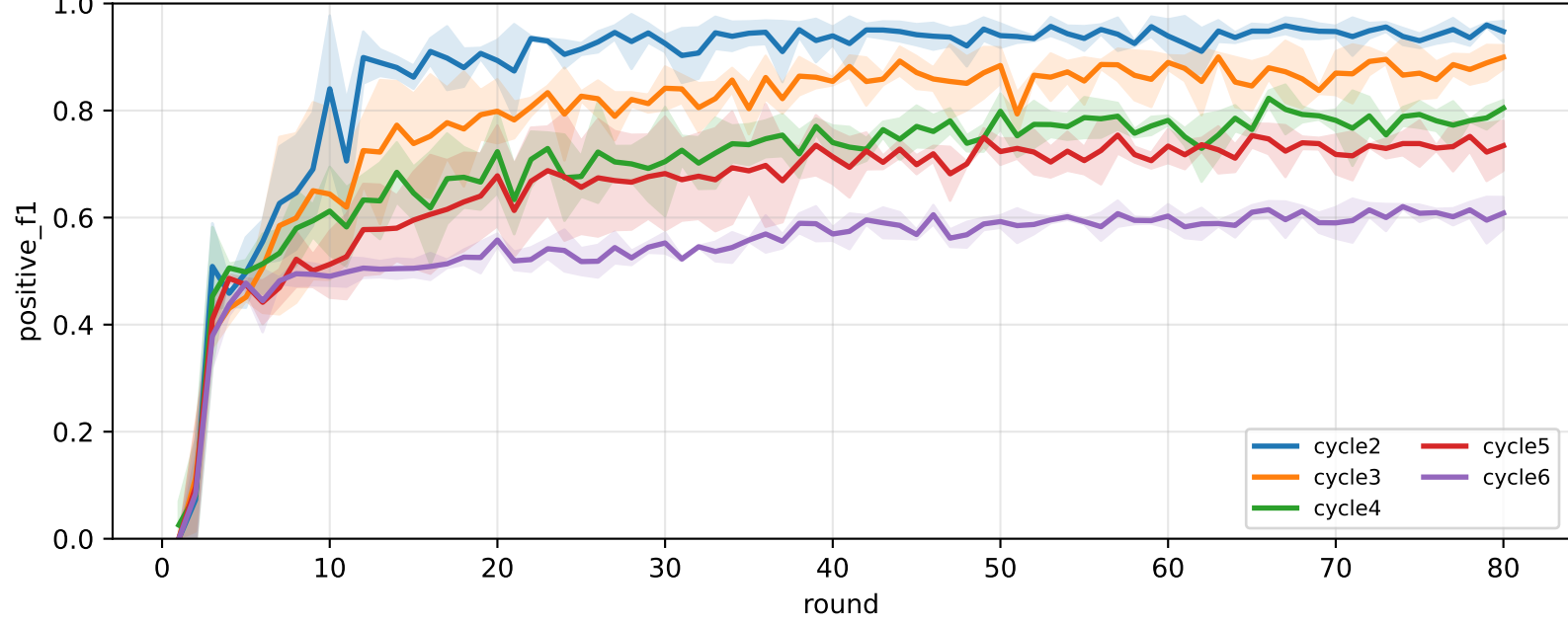
Pooled local centralized positive F1



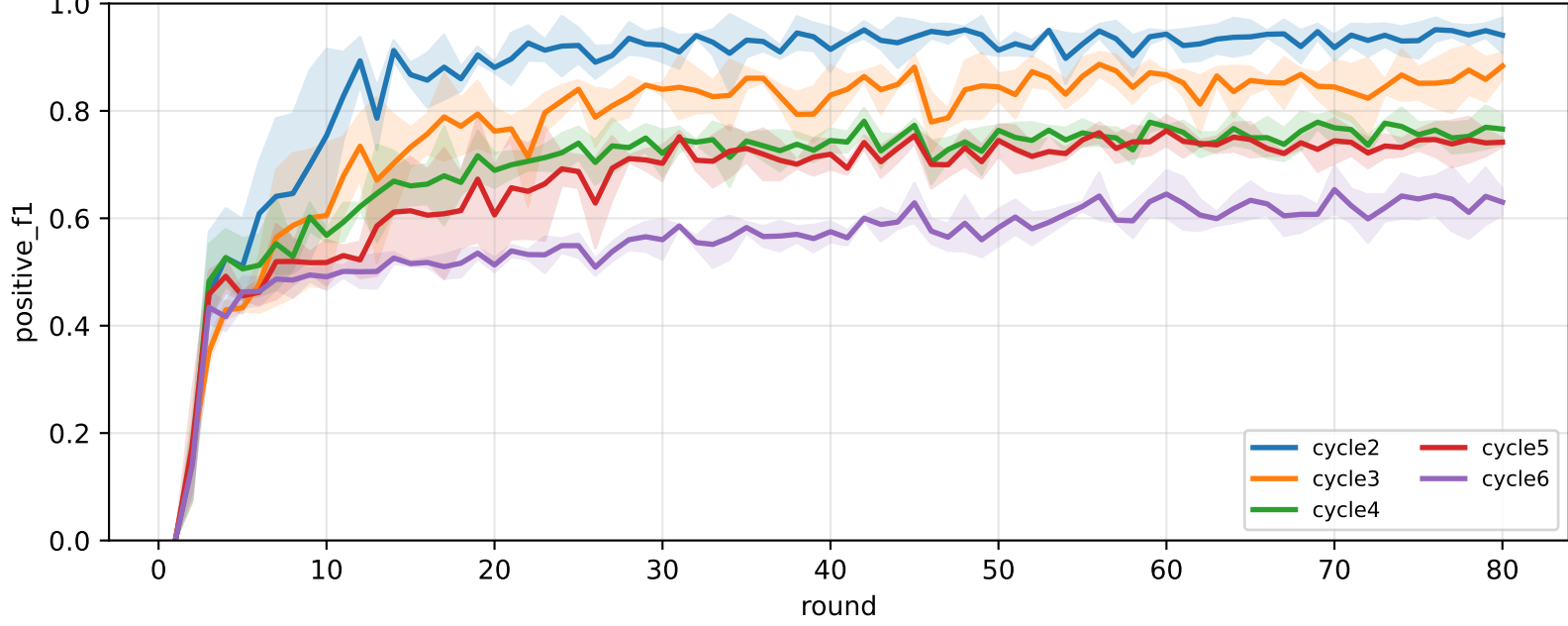
Pooled FedAvg positive F1



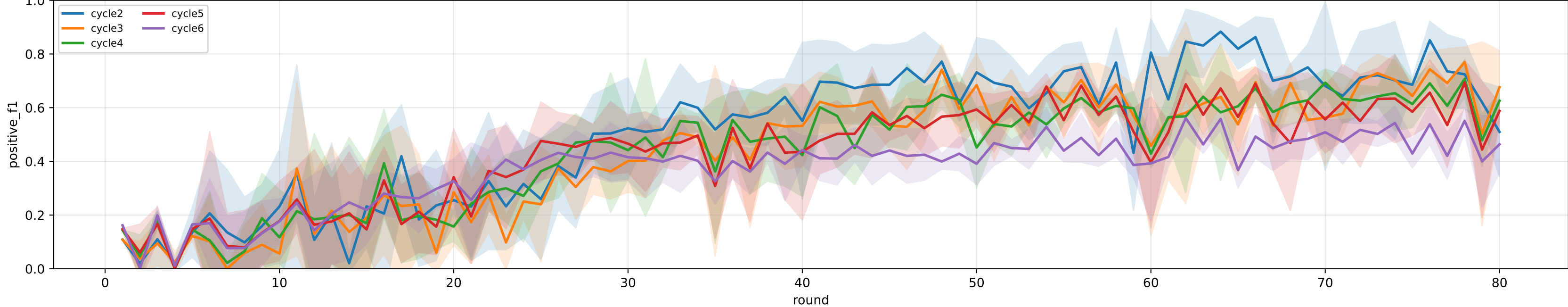
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1



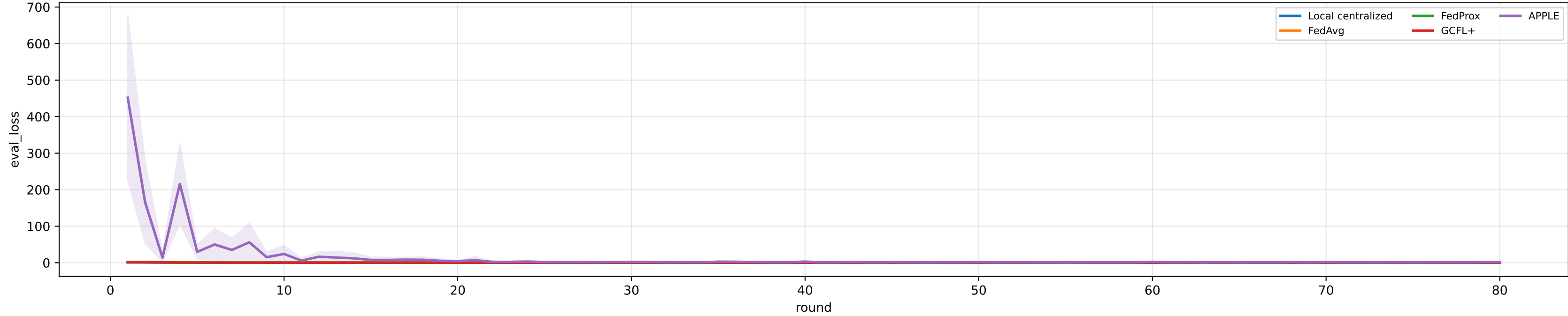
Controlled task-specialization level: High | family aggregated / pooled performance |  
family=specialized\_cycle2 | n\_clients=1 |  
model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64

family=specialized\_cycle2 | n\_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |  
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |  
specialization=high | specialization\_fraction=0.80 | global\_visible\_support=0.36 | task\_jsd\_mean=0.162 | task\_jsd\_median=0.162 | task\_jsd\_max=0.165 |  
family\_centroid\_jsd\_mean=0.162

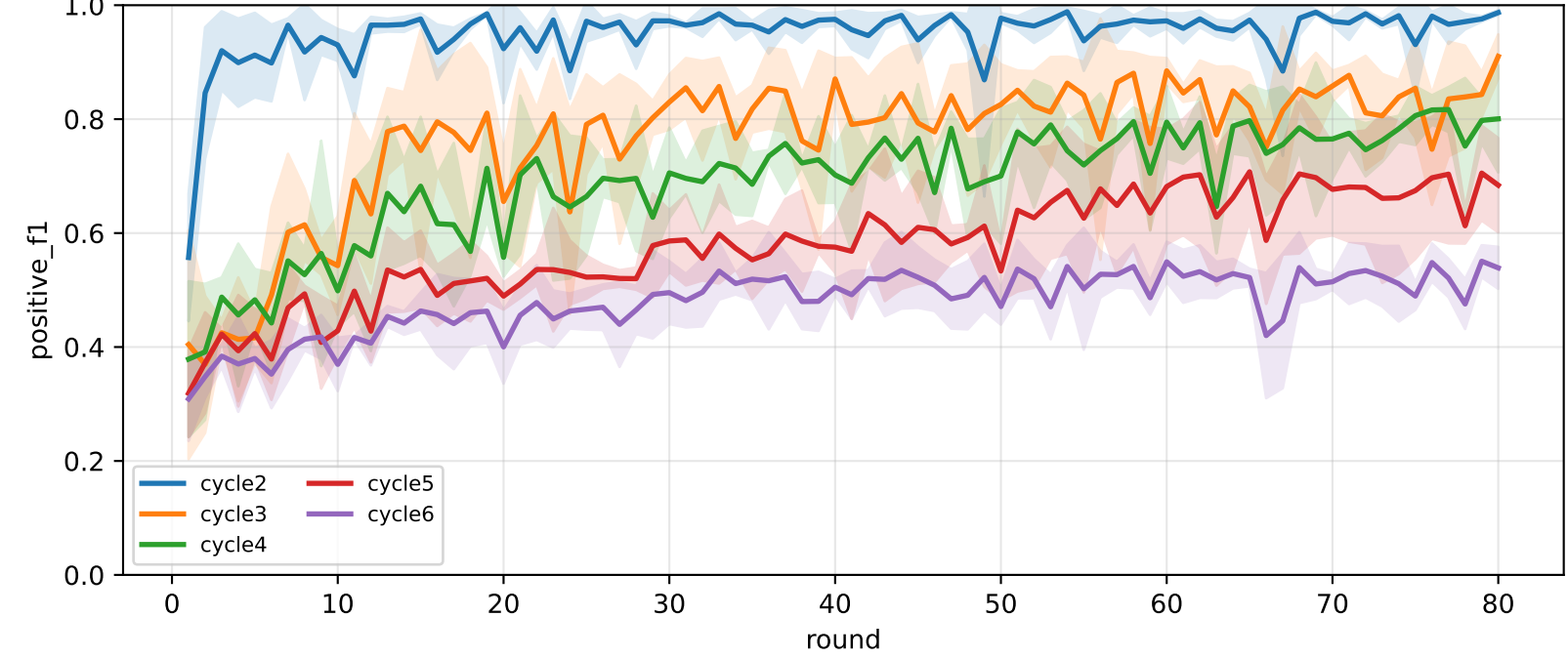
Best logged test performance (mean ± std over seeds)

| Algorithm         | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Local centralized | 71.8% ± 2.2%        | 76.9% ± 0.8%        | <b>98.1% ± 0.8%</b> | 90.4% ± 2.4%        | 76.9% ± 4.5%        | 65.2% ± 5.3%        | 53.7% ± 4.1%        |
| FedAvg            | 77.0% ± 1.8%        | 80.7% ± 1.8%        | 95.6% ± 0.6%        | 92.2% ± 3.4%        | 81.6% ± 1.2%        | <b>74.7% ± 2.9%</b> | 59.3% ± 1.1%        |
| FedProx           | 76.0% ± 1.5%        | 80.1% ± 1.3%        | 95.9% ± 1.6%        | <b>92.6% ± 2.2%</b> | <b>83.0% ± 2.5%</b> | 70.1% ± 3.3%        | 58.8% ± 2.0%        |
| GCFL+             | <b>78.3% ± 1.7%</b> | <b>80.9% ± 1.3%</b> | 95.6% ± 0.7%        | 90.4% ± 1.2%        | 79.1% ± 2.1%        | 73.9% ± 0.4%        | <b>65.6% ± 3.8%</b> |
| APPLE             | 54.4% ± 4.4%        | 61.0% ± 2.2%        | 86.5% ± 6.1%        | 69.3% ± 7.4%        | 56.4% ± 6.7%        | 53.0% ± 10.4%       | 39.7% ± 5.3%        |

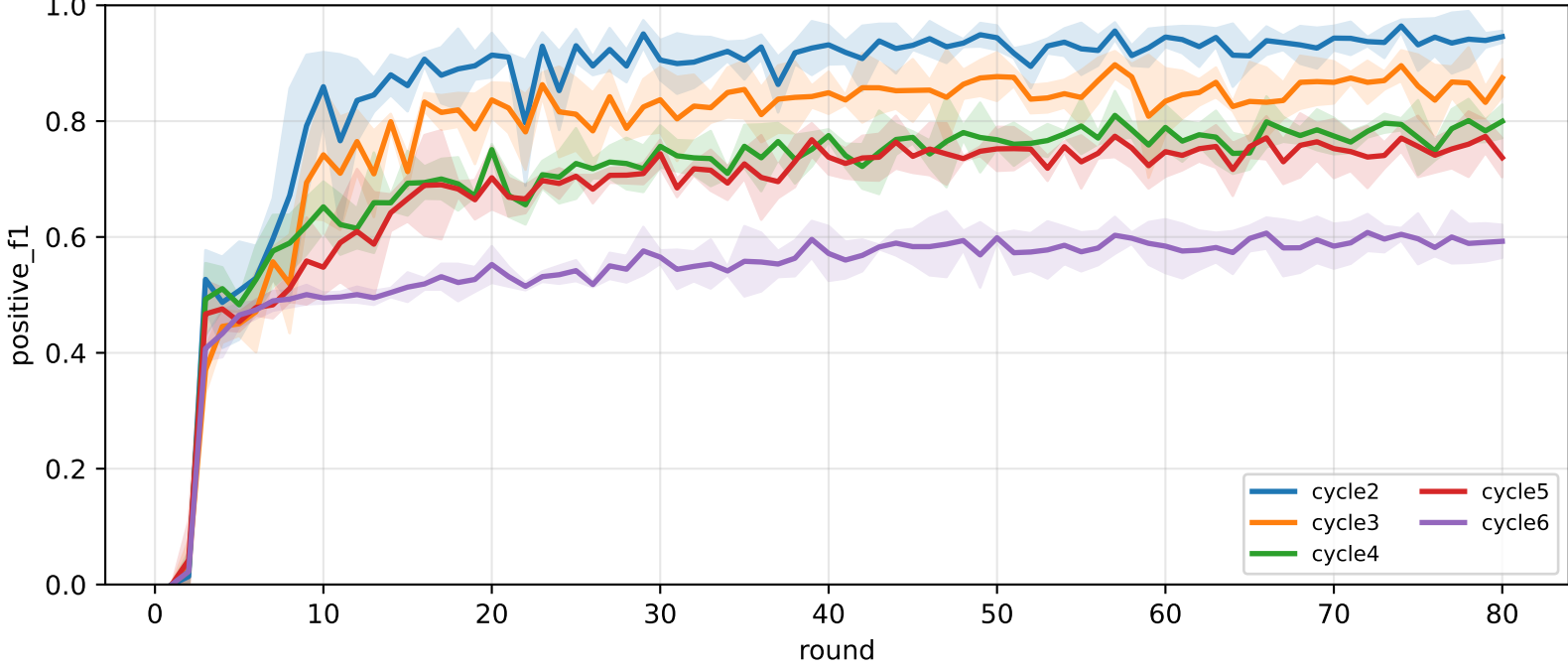
Validation loss comparison



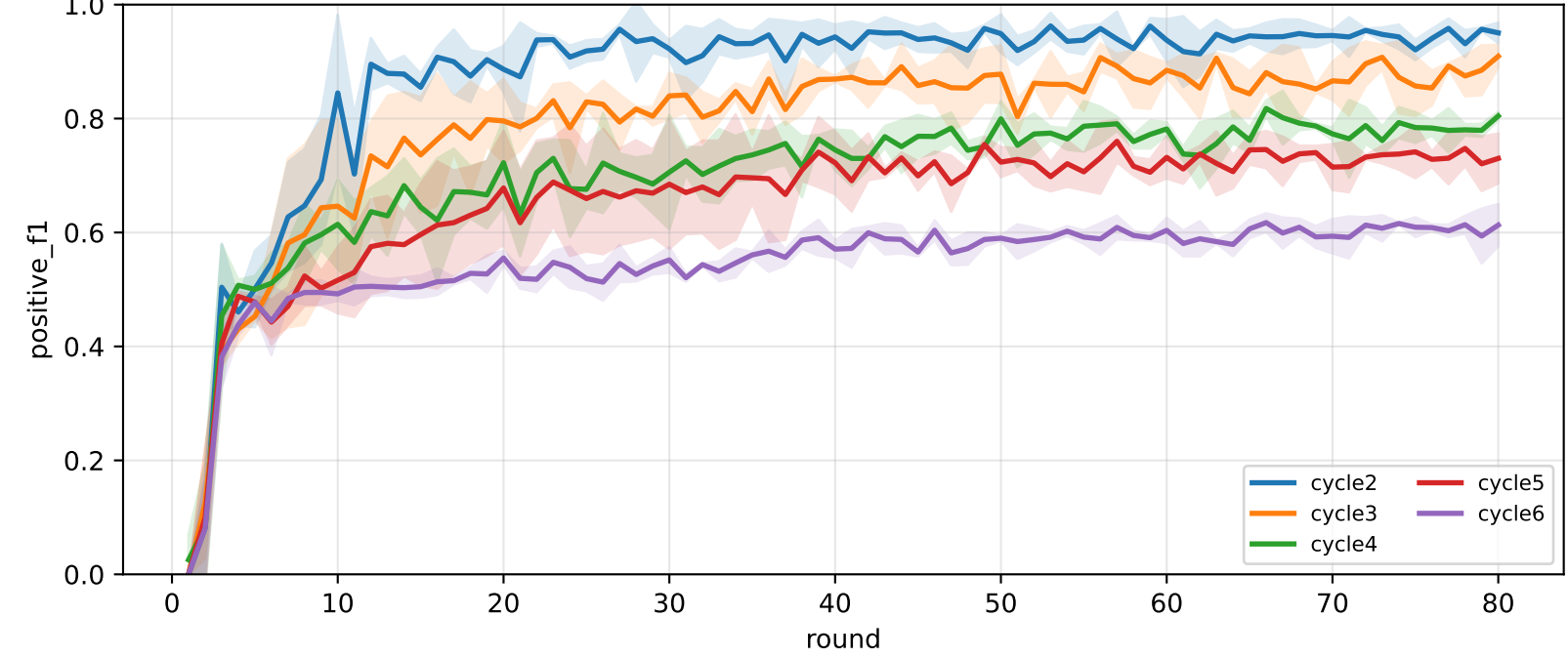
Pooled local centralized positive F1



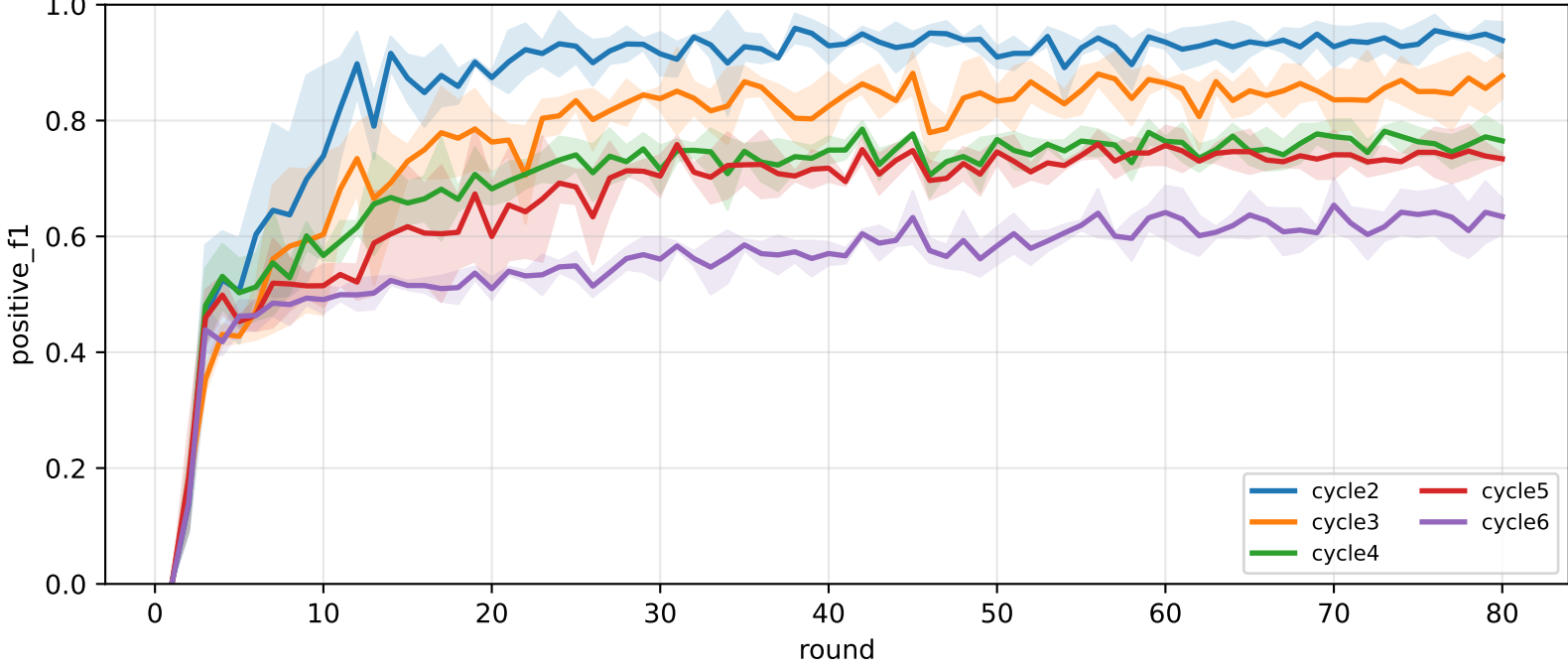
Pooled FedAvg positive F1



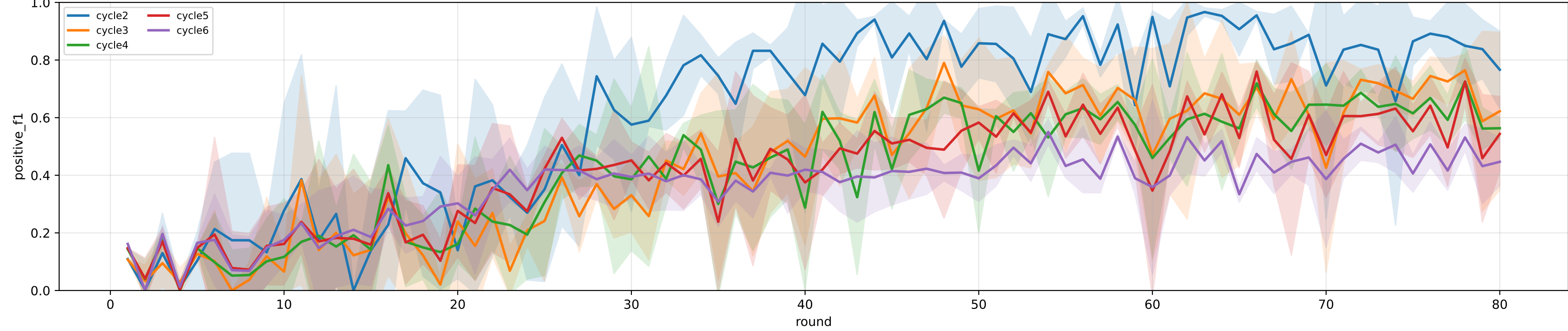
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1





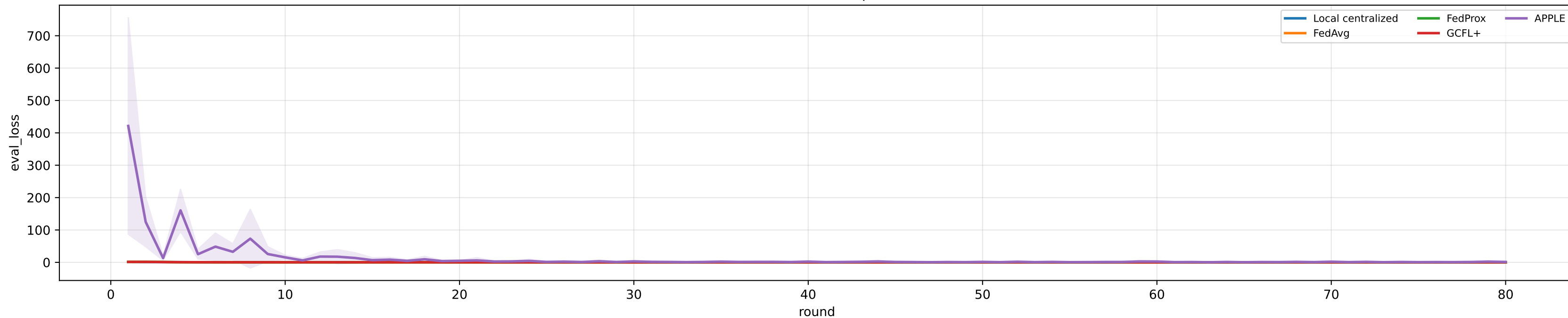
Controlled task-specialization level: High | family aggregated / pooled performance |  
family=specialized\_cycle3 | n\_clients=1 |  
model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64

family=specialized\_cycle3 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |  
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |  
specialization=high | specialization\_fraction=0.80 | global\_visible\_support=0.36 | task\_jsd\_mean=0.162 | task\_jsd\_median=0.162 | task\_jsd\_max=0.165 |  
family\_centroid\_jsd\_mean=0.162

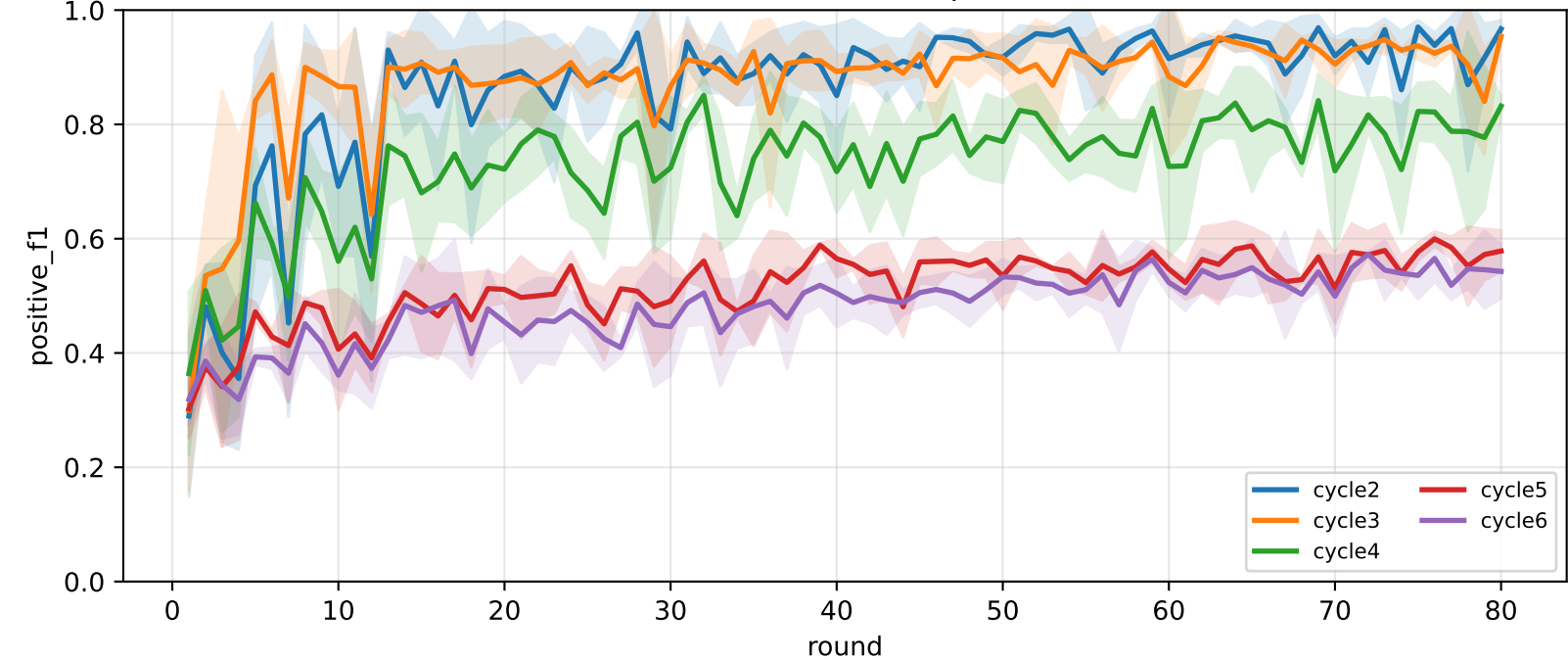
Best logged test performance (mean ± std over seeds)

| Algorithm         | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Local centralized | 69.6% ± 3.7%        | 76.6% ± 3.0%        | <b>96.4% ± 1.1%</b> | <b>94.7% ± 1.9%</b> | <b>86.7% ± 5.4%</b> | 52.7% ± 5.2%        | 52.7% ± 4.1%        |
| FedAvg            | 76.9% ± 1.3%        | <b>80.7% ± 1.4%</b> | 94.9% ± 1.8%        | 93.0% ± 2.3%        | 82.0% ± 0.5%        | <b>74.4% ± 2.1%</b> | 59.0% ± 1.0%        |
| FedProx           | 76.2% ± 1.7%        | 80.2% ± 1.1%        | 95.1% ± 0.8%        | 93.3% ± 2.4%        | 83.6% ± 2.2%        | 70.0% ± 4.8%        | 59.2% ± 2.6%        |
| GCFL+             | <b>78.0% ± 2.0%</b> | <b>80.7% ± 1.5%</b> | 95.6% ± 0.3%        | 90.8% ± 1.6%        | 79.5% ± 1.5%        | 72.8% ± 1.4%        | <b>65.0% ± 3.7%</b> |
| APPLE             | 59.0% ± 3.7%        | 63.7% ± 2.5%        | 75.8% ± 2.7%        | 81.0% ± 10.9%       | 59.2% ± 6.4%        | 58.6% ± 9.1%        | 43.9% ± 4.6%        |

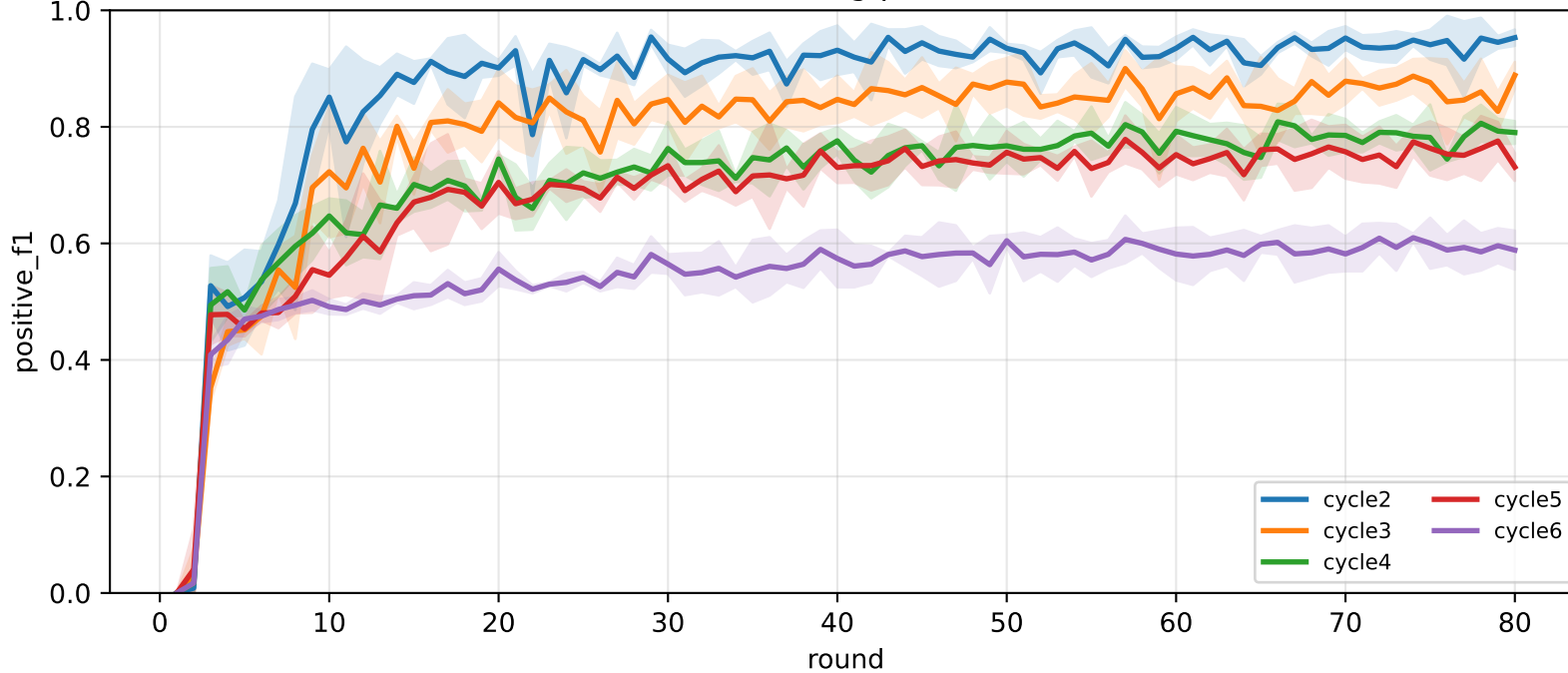
Validation loss comparison



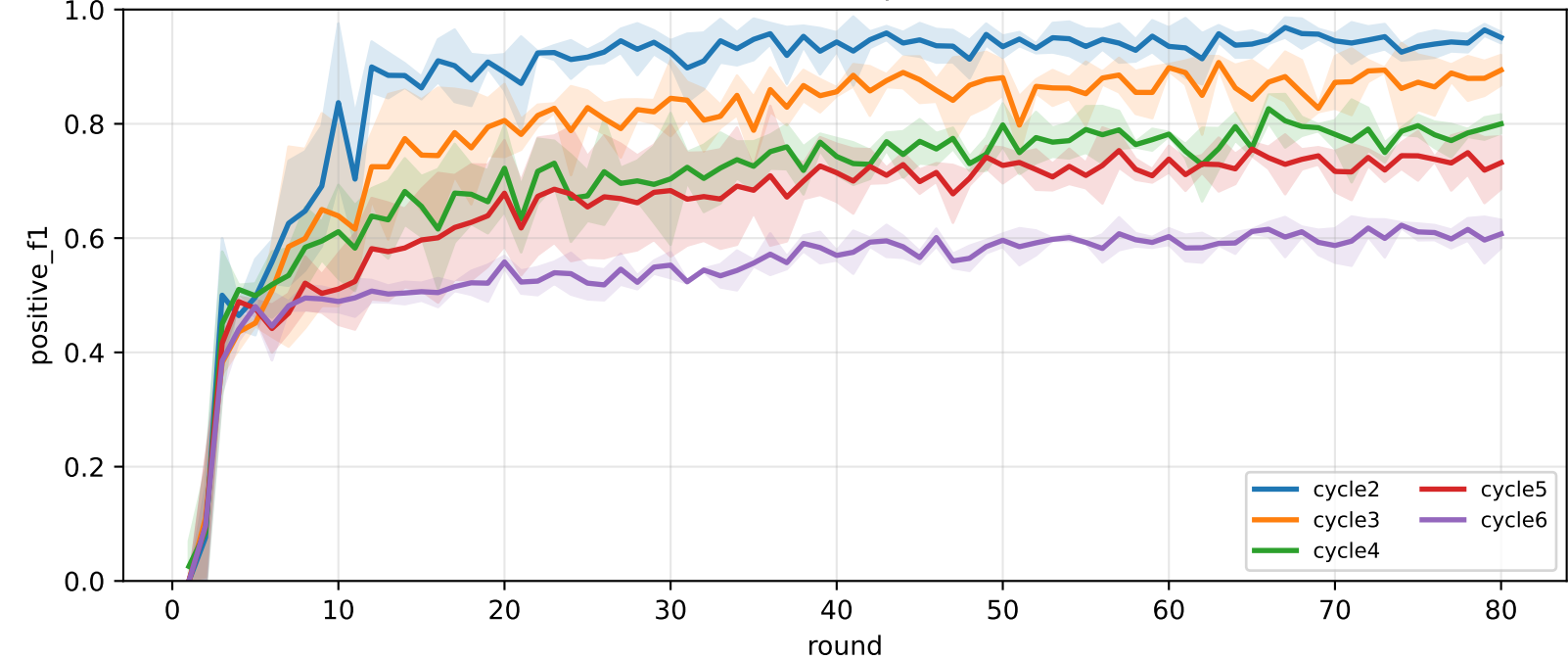
Pooled local centralized positive F1



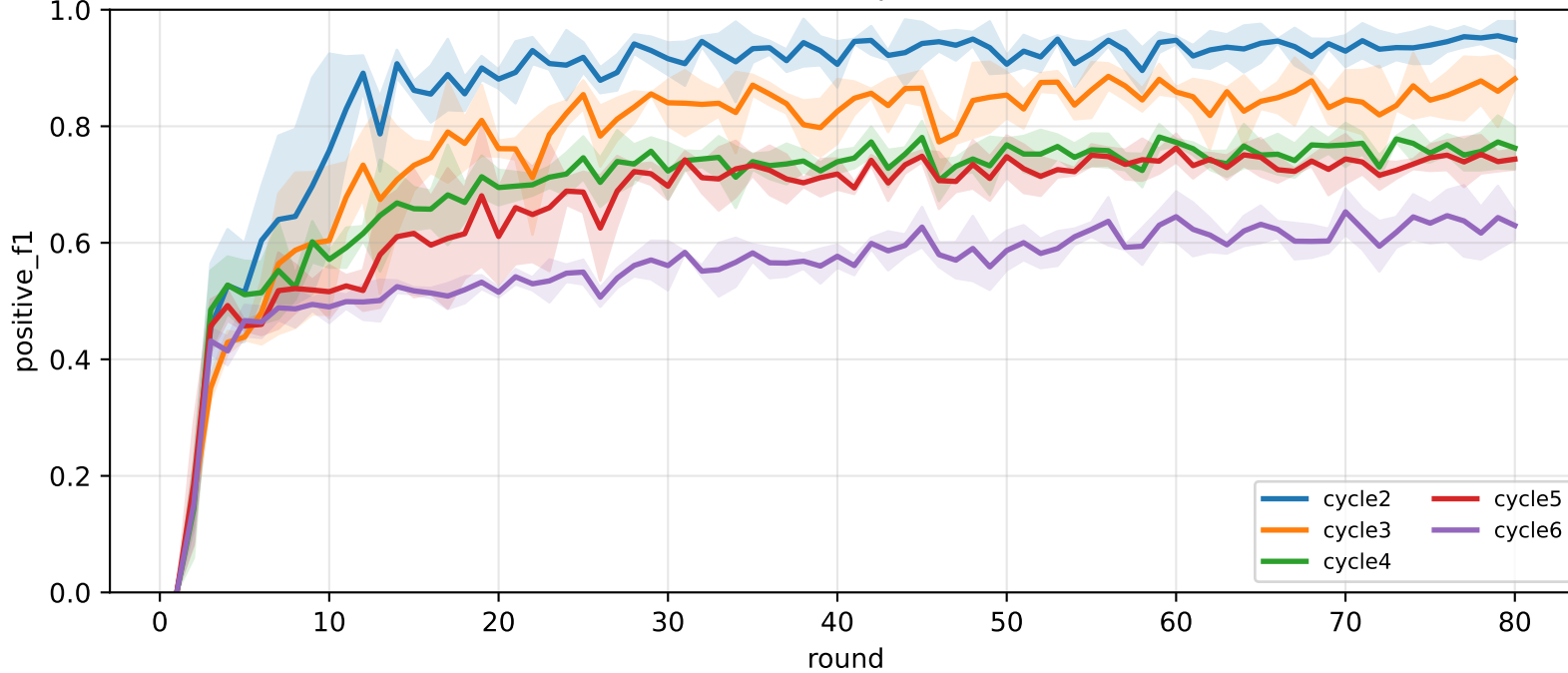
Pooled FedAvg positive F1



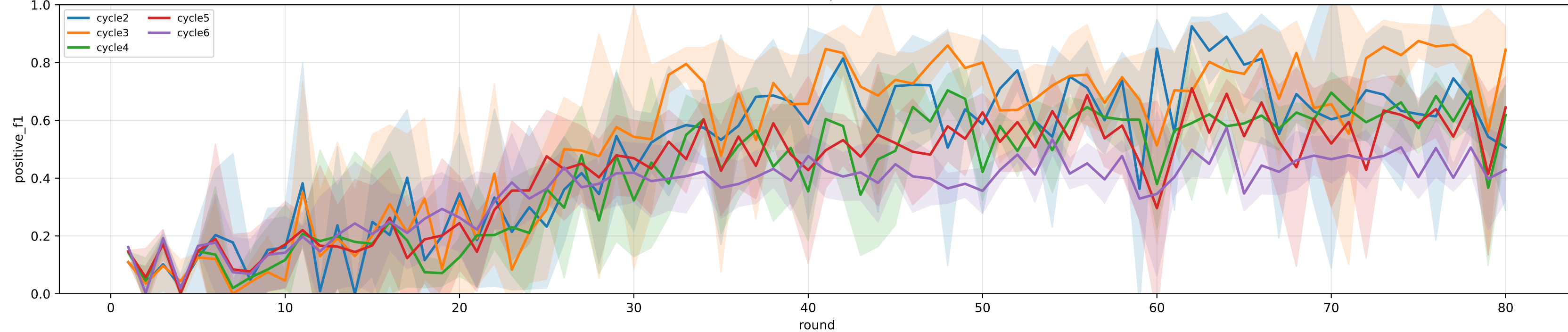
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1



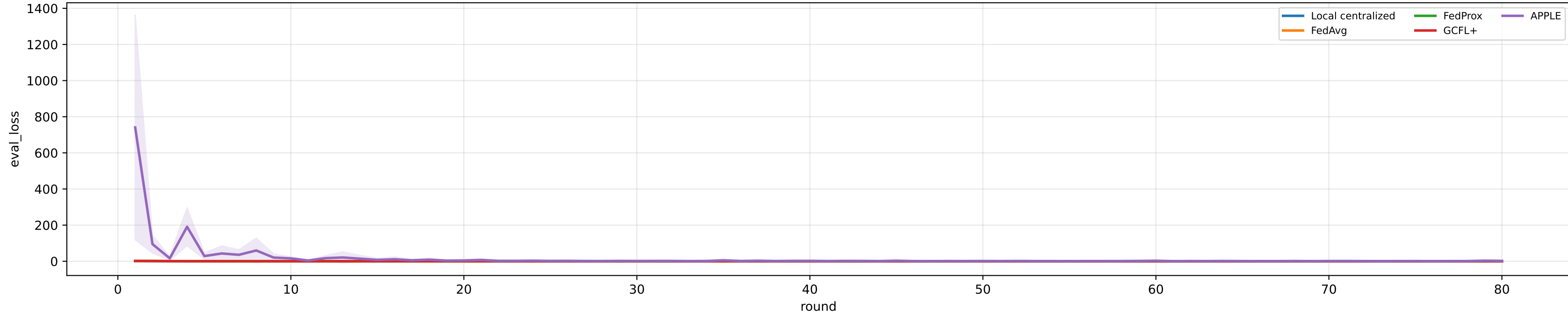
Controlled task-specialization level: High | family aggregated / pooled performance |  
family=specialized\_cycle4 | n\_clients=1 |  
model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64

family=specialized\_cycle4 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% |  
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |  
specialization=high | specialization\_fraction=0.80 | global\_visible\_support=0.36 | task\_jsd\_mean=0.162 | task\_jsd\_median=0.162 | task\_jsd\_max=0.165 |  
family\_centroid\_jsd\_mean=0.162

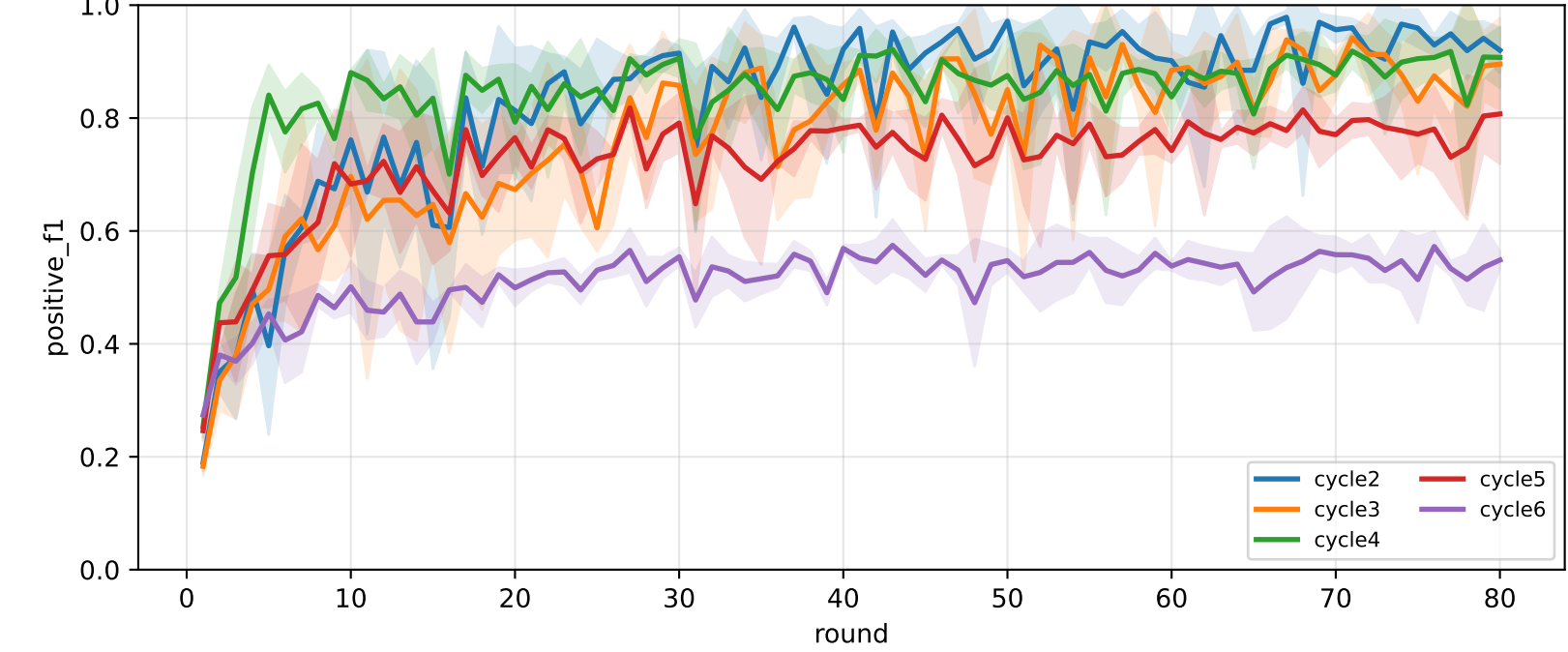
Best logged test performance (mean ± std over seeds)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 78.0% ± 1.1% | 82.8% ± 1.1% | 95.4% ± 2.6%  | 95.6% ± 1.0%  | 91.0% ± 2.8%  | 76.7% ± 2.4%  | 55.2% ± 0.7%  |
| FedAvg            | 76.7% ± 1.9% | 80.5% ± 1.9% | 95.2% ± 0.1%  | 92.3% ± 3.6%  | 81.3% ± 1.8%  | 74.7% ± 3.4%  | 59.0% ± 1.3%  |
| FedProx           | 75.8% ± 1.4% | 79.7% ± 0.9% | 95.2% ± 1.3%  | 92.1% ± 2.4%  | 83.3% ± 1.0%  | 69.5% ± 3.4%  | 58.6% ± 2.2%  |
| GCFL+             | 77.4% ± 2.1% | 80.0% ± 1.5% | 95.1% ± 0.6%  | 89.1% ± 0.4%  | 78.6% ± 2.3%  | 71.9% ± 1.5%  | 65.2% ± 4.2%  |
| APPLE             | 58.3% ± 4.0% | 62.6% ± 2.9% | 73.8% ± 15.3% | 69.4% ± 11.8% | 67.7% ± 5.4%  | 58.8% ± 7.0%  | 43.2% ± 5.7%  |

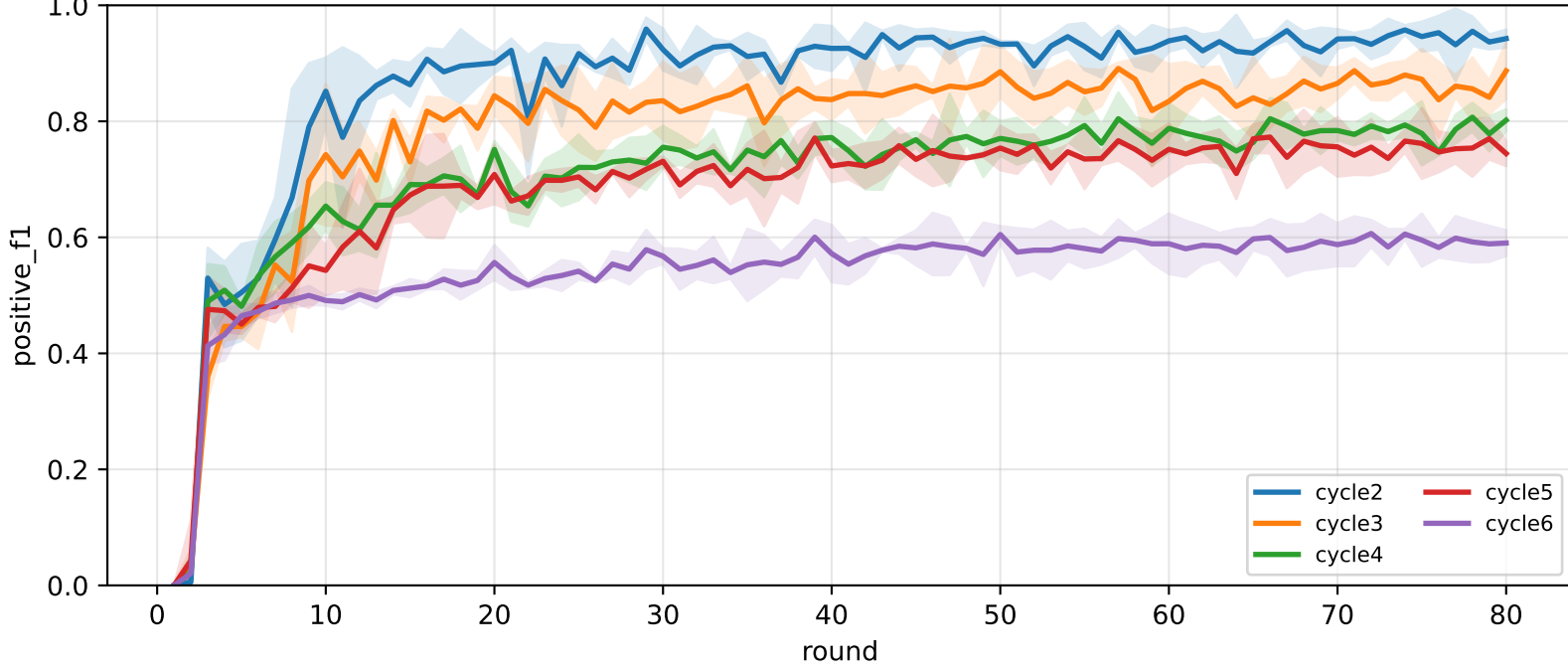
Validation loss comparison



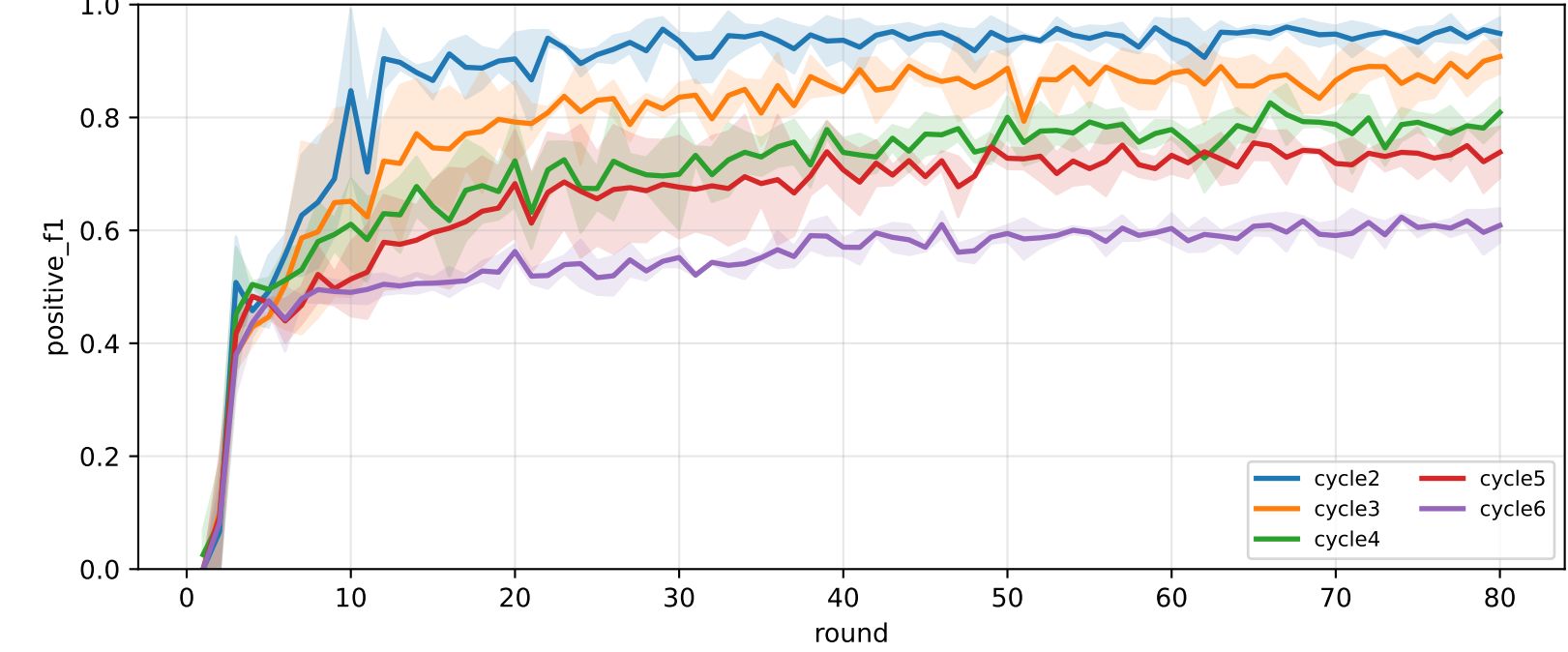
Pooled local centralized positive F1



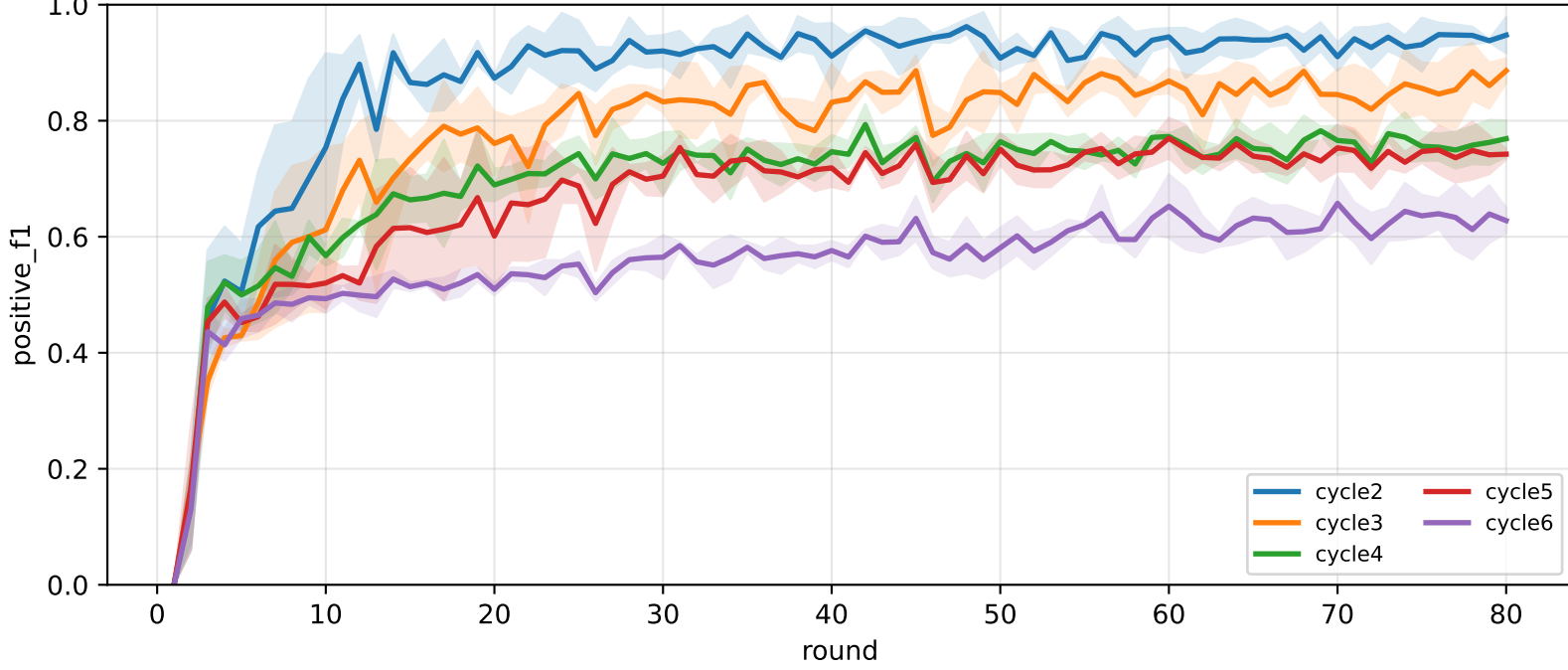
Pooled FedAvg positive F1



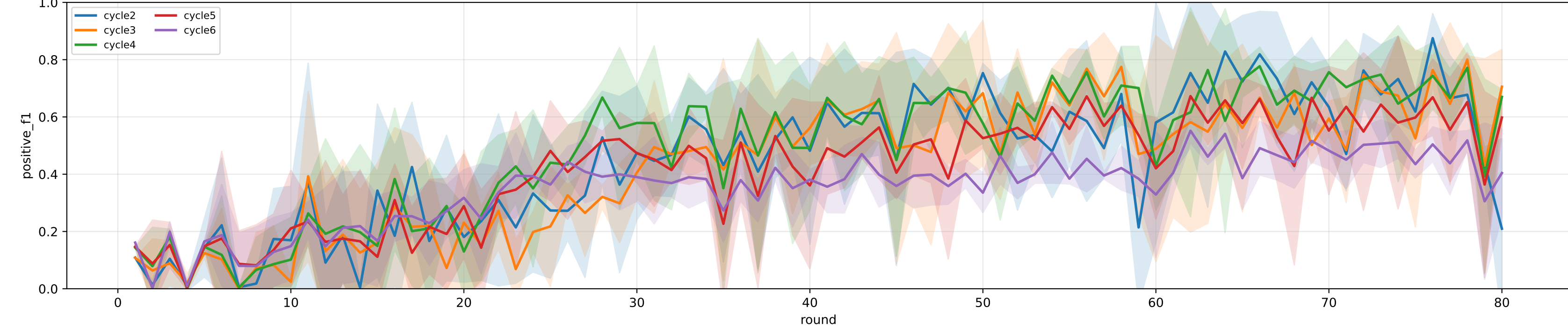
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1





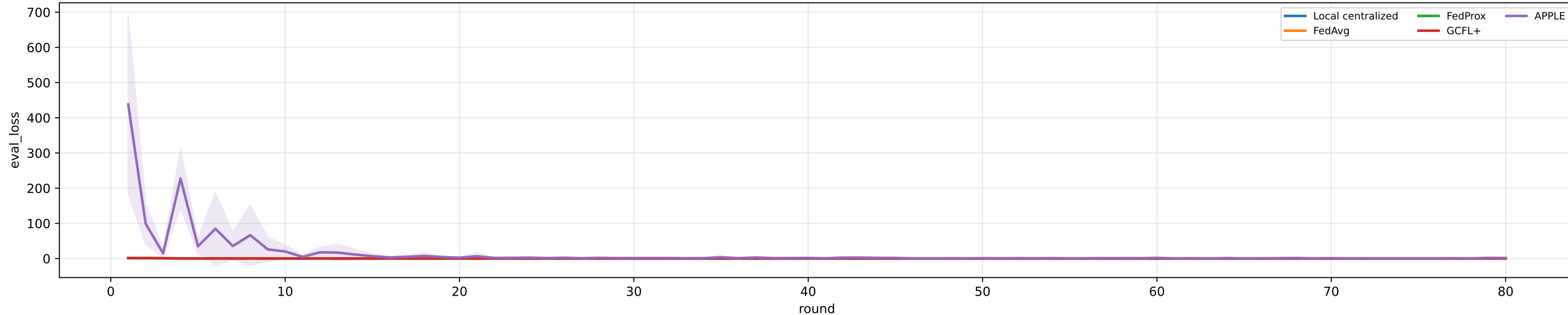
Controlled task-specialization level: High | family aggregated / pooled performance |  
family=specialized\_cycle5 | n\_clients=1 |  
model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64

family=specialized\_cycle5 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |  
cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |  
specialization=high | specialization\_fraction=0.80 | global\_visible\_support=0.36 | task\_jsd\_mean=0.162 | task\_jsd\_median=0.162 | task\_jsd\_max=0.165 |  
family\_centroid\_jsd\_mean=0.162

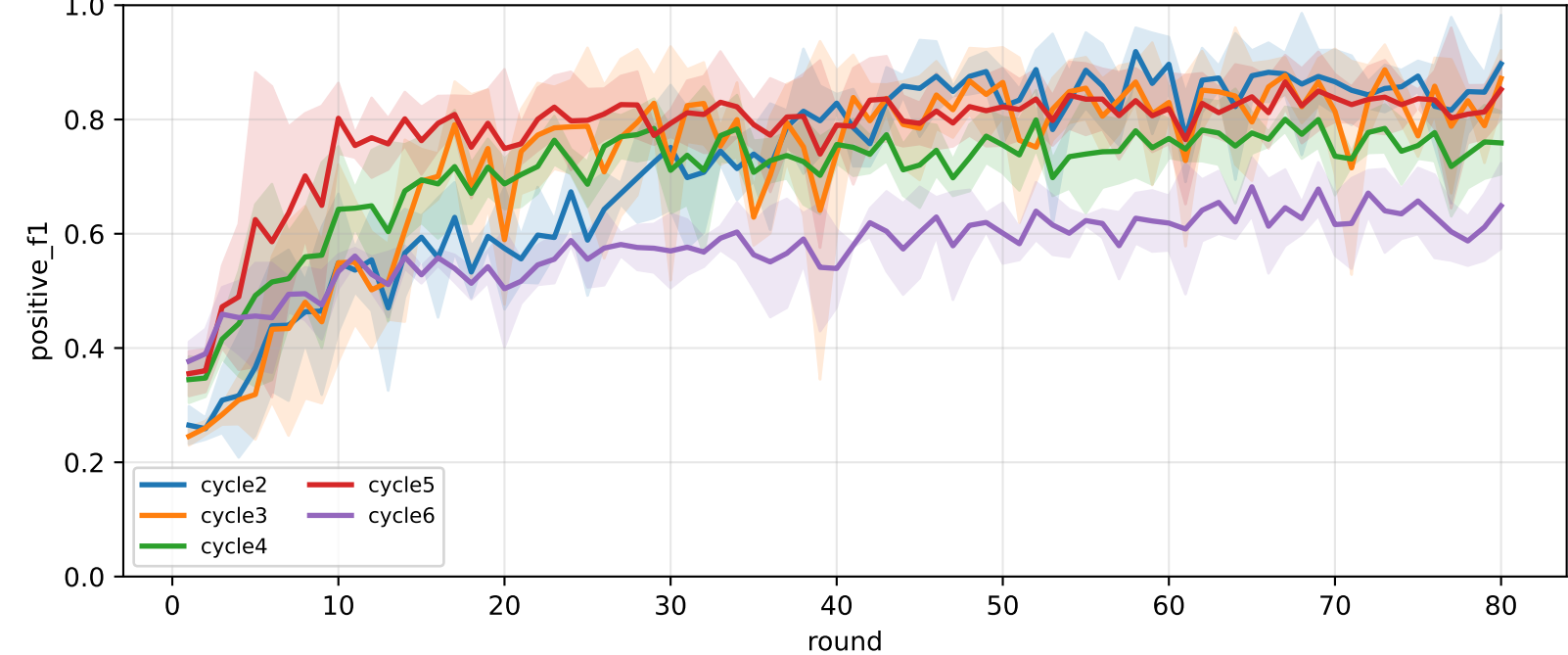
Best logged test performance (mean ± std over seeds)

| Algorithm         | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Local centralized | 75.8% ± 0.9%        | 78.3% ± 1.5%        | 87.0% ± 1.3%        | 87.4% ± 3.9%        | 77.0% ± 3.6%        | <b>79.2% ± 5.6%</b> | 60.7% ± 3.5%        |
| FedAvg            | 76.5% ± 1.9%        | 80.2% ± 2.0%        | 94.7% ± 0.9%        | 91.3% ± 3.3%        | 81.3% ± 1.5%        | 74.4% ± 3.6%        | 59.0% ± 1.0%        |
| FedProx           | 75.7% ± 1.5%        | 79.5% ± 1.3%        | 93.2% ± 1.4%        | <b>91.8% ± 2.7%</b> | <b>84.0% ± 1.6%</b> | 69.5% ± 3.3%        | 58.9% ± 2.3%        |
| GCFL+             | <b>77.8% ± 1.2%</b> | <b>80.5% ± 0.7%</b> | <b>95.5% ± 0.8%</b> | 90.2% ± 0.8%        | 78.5% ± 2.2%        | 73.7% ± 1.4%        | <b>64.4% ± 3.2%</b> |
| APPLE             | 56.9% ± 6.4%        | 61.6% ± 5.3%        | 73.4% ± 11.1%       | 69.6% ± 11.5%       | 60.5% ± 9.5%        | 64.0% ± 8.1%        | 40.7% ± 7.9%        |

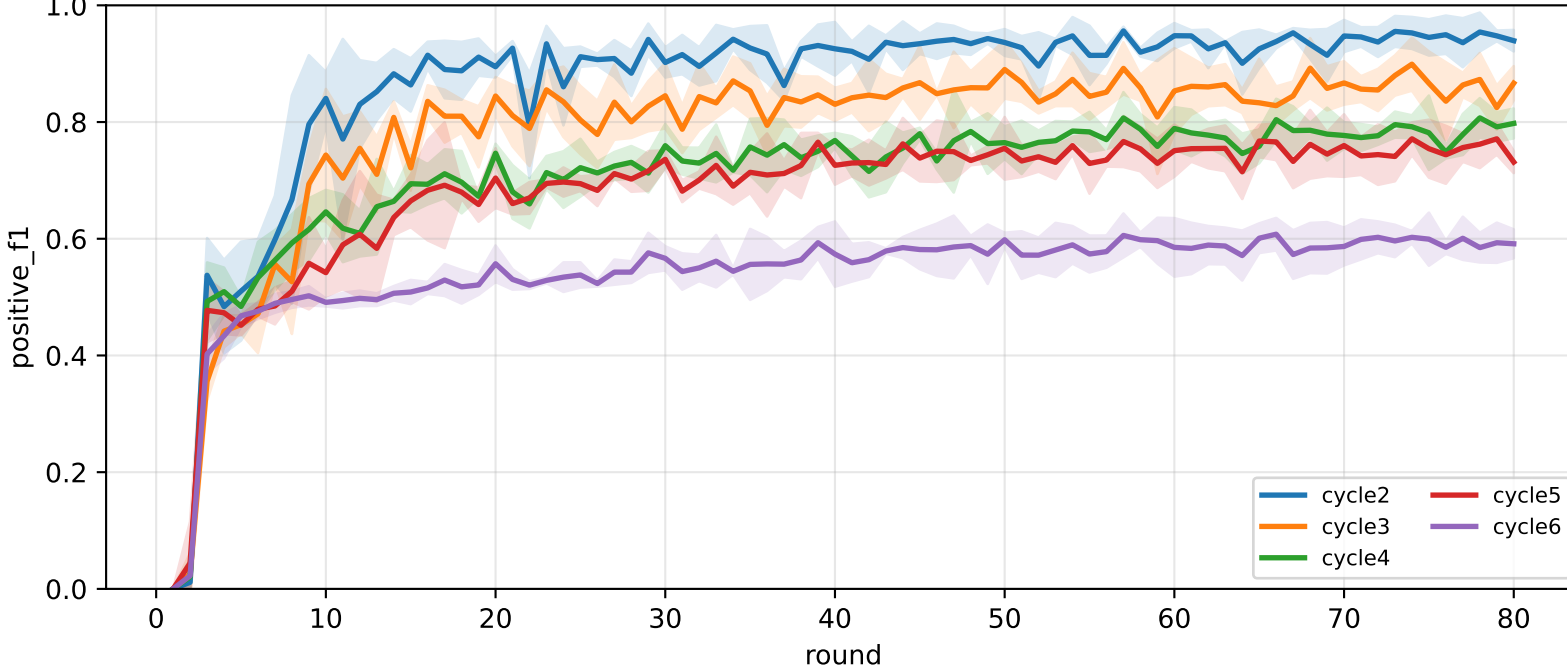
Validation loss comparison



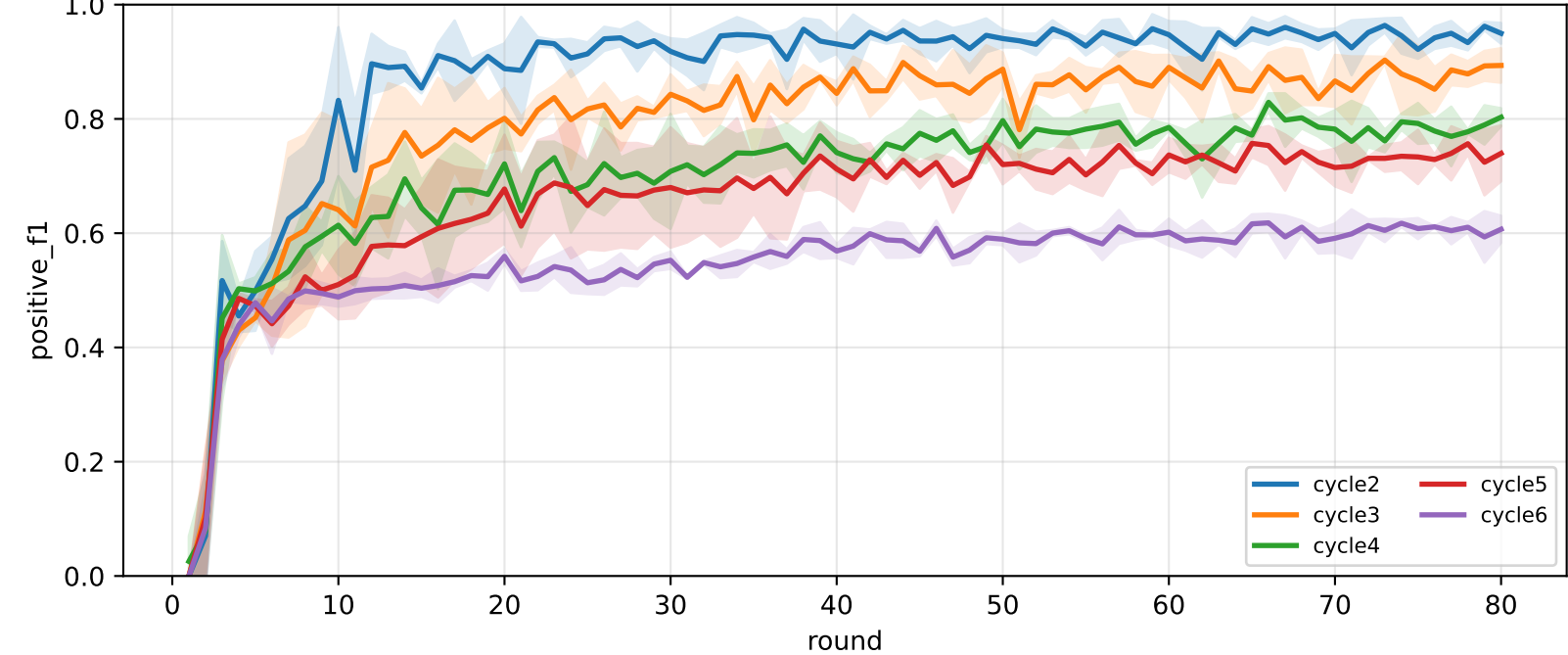
Pooled local centralized positive F1



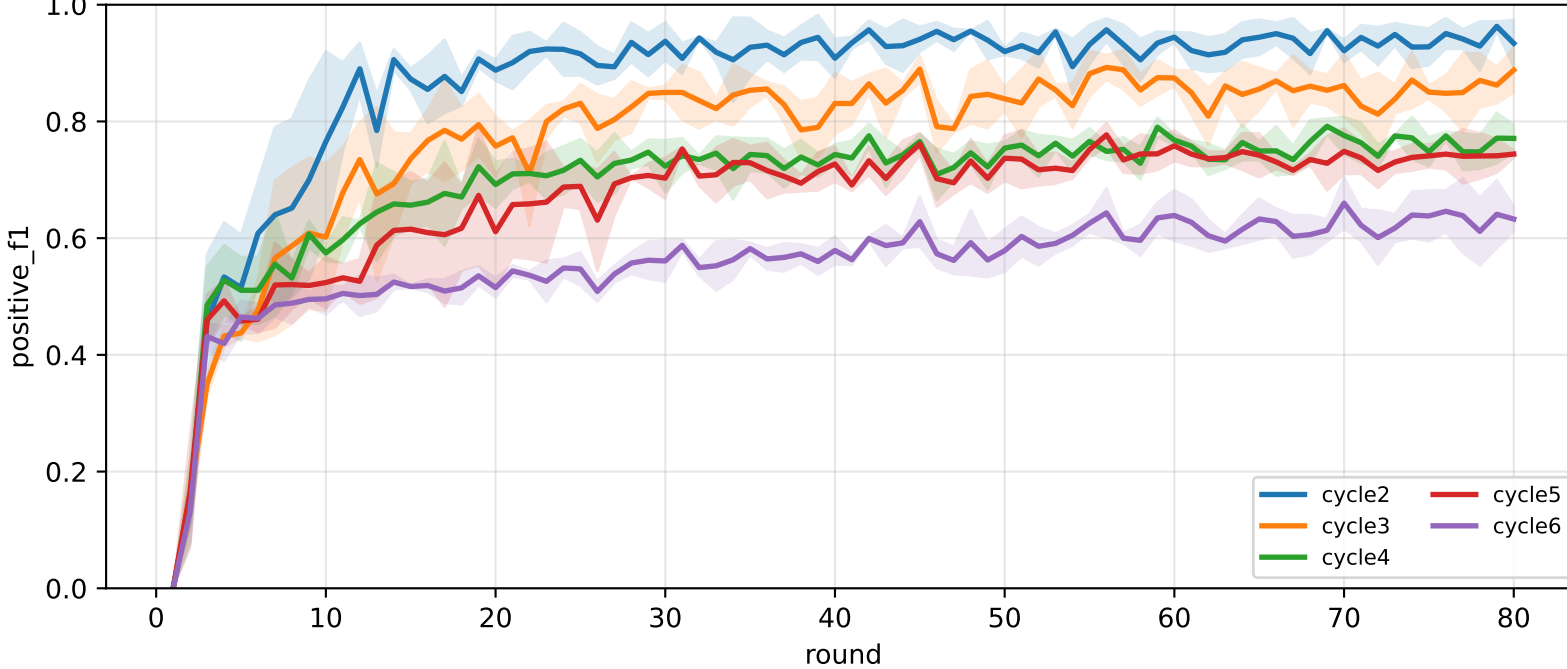
Pooled FedAvg positive F1



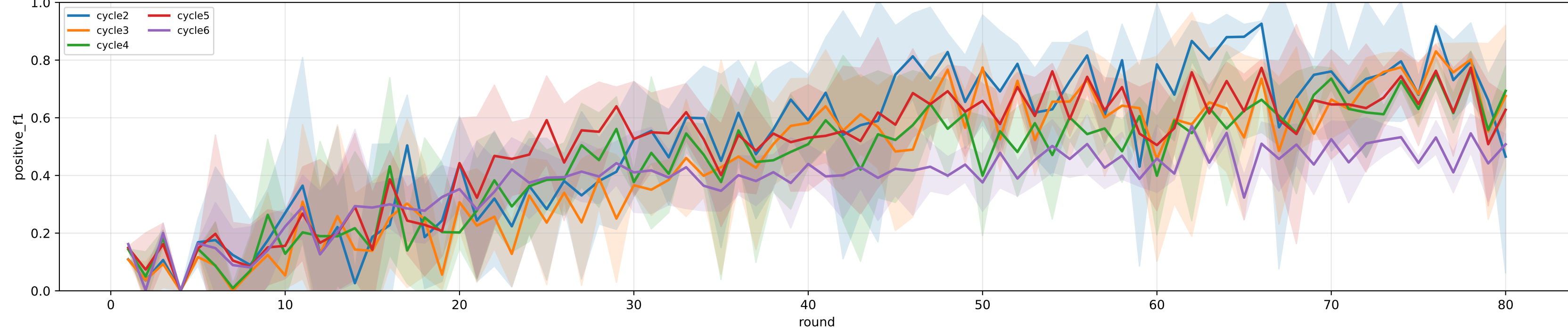
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1



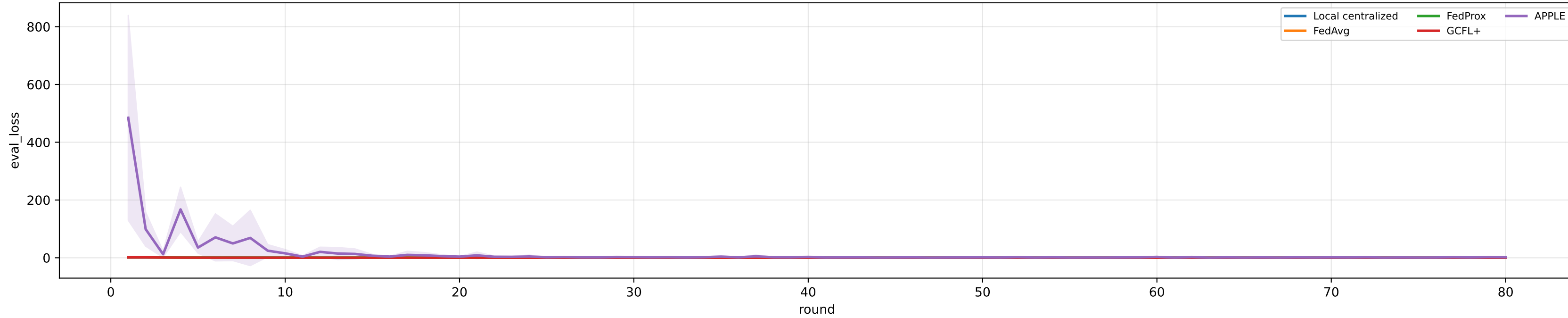
Controlled task-specialization level: High | family aggregated / pooled performance |  
family=specialized\_cycle6 | n\_clients=1 |  
model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64

family=specialized\_cycle6 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |  
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3% |  
specialization=high | specialization\_fraction=0.80 | global\_visible\_support=0.36 | task\_jsd\_mean=0.162 | task\_jsd\_median=0.162 | task\_jsd\_max=0.165 |  
family\_centroid\_jsd\_mean=0.162

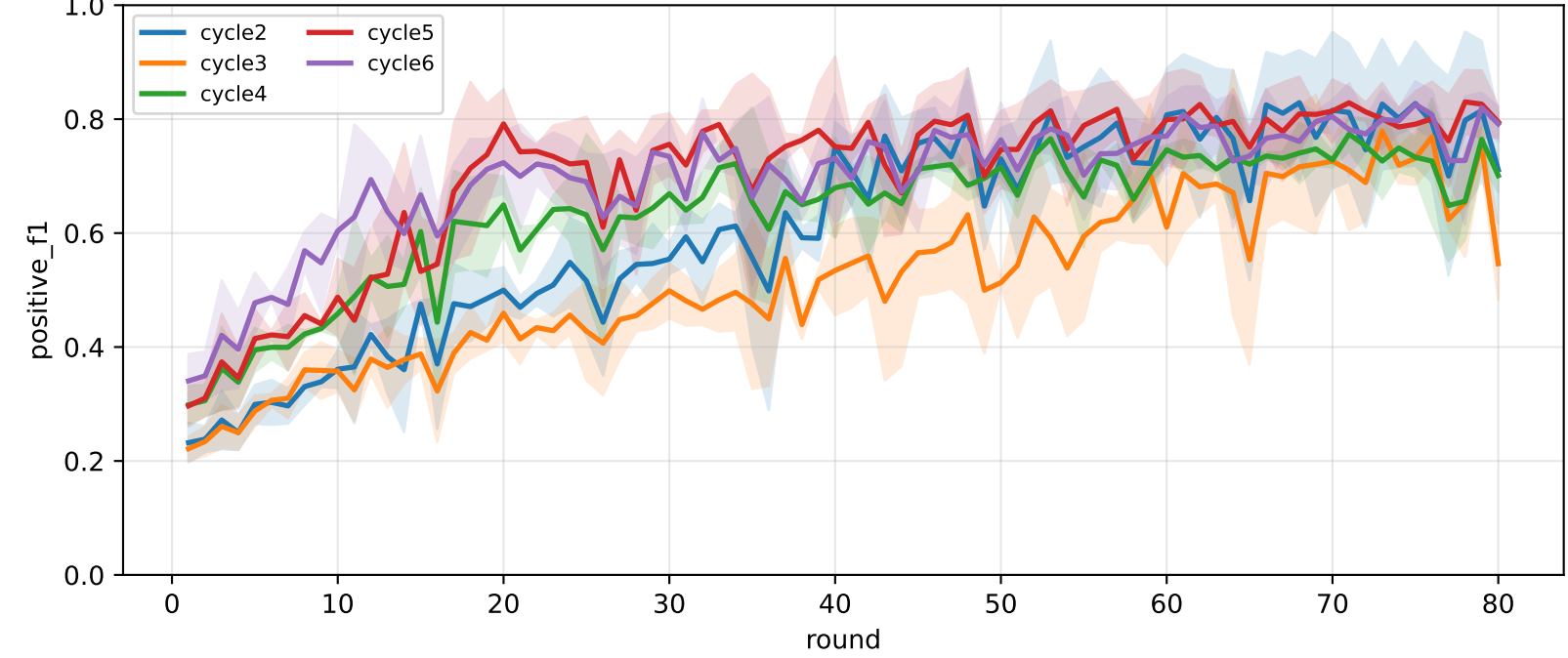
Best logged test performance (mean ± std over seeds)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 78.5% ± 3.5% | 79.1% ± 2.8% | 83.3% ± 8.5%  | 79.7% ± 5.3%  | 76.3% ± 4.2%  | 77.7% ± 6.0%  | 78.4% ± 6.9%  |
| FedAvg            | 77.0% ± 2.2% | 80.7% ± 2.4% | 95.2% ± 3.0%  | 92.8% ± 2.9%  | 81.7% ± 2.4%  | 75.0% ± 3.6%  | 58.9% ± 1.0%  |
| FedProx           | 76.4% ± 1.2% | 80.5% ± 0.8% | 95.4% ± 0.9%  | 93.5% ± 2.0%  | 83.9% ± 1.1%  | 70.7% ± 3.1%  | 58.9% ± 1.9%  |
| GCFL+             | 77.5% ± 1.6% | 80.2% ± 1.4% | 95.8% ± 1.7%  | 90.5% ± 1.3%  | 77.9% ± 1.9%  | 72.1% ± 0.3%  | 64.9% ± 3.1%  |
| APPLE             | 57.3% ± 3.0% | 59.1% ± 2.2% | 71.9% ± 6.2%  | 59.6% ± 8.7%  | 54.0% ± 4.1%  | 55.6% ± 8.7%  | 54.6% ± 3.9%  |

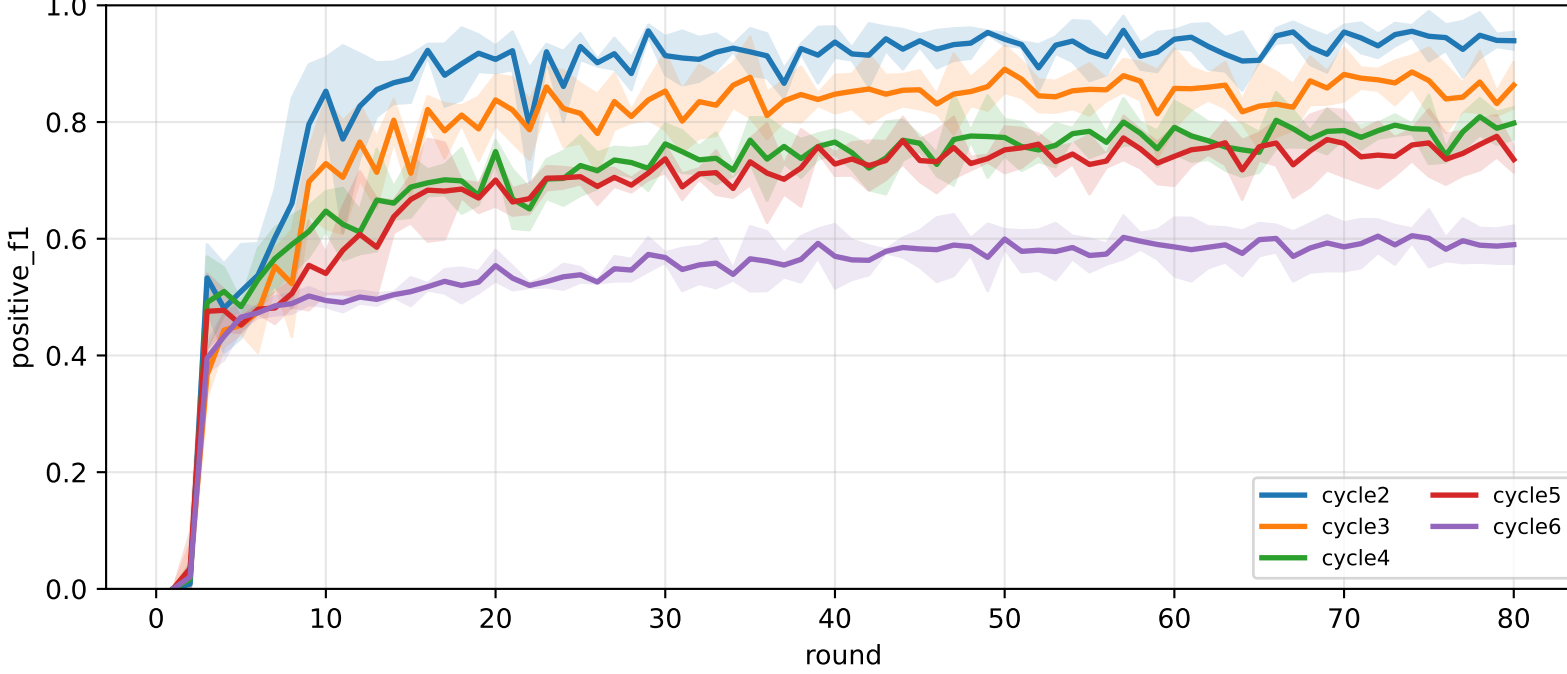
Validation loss comparison



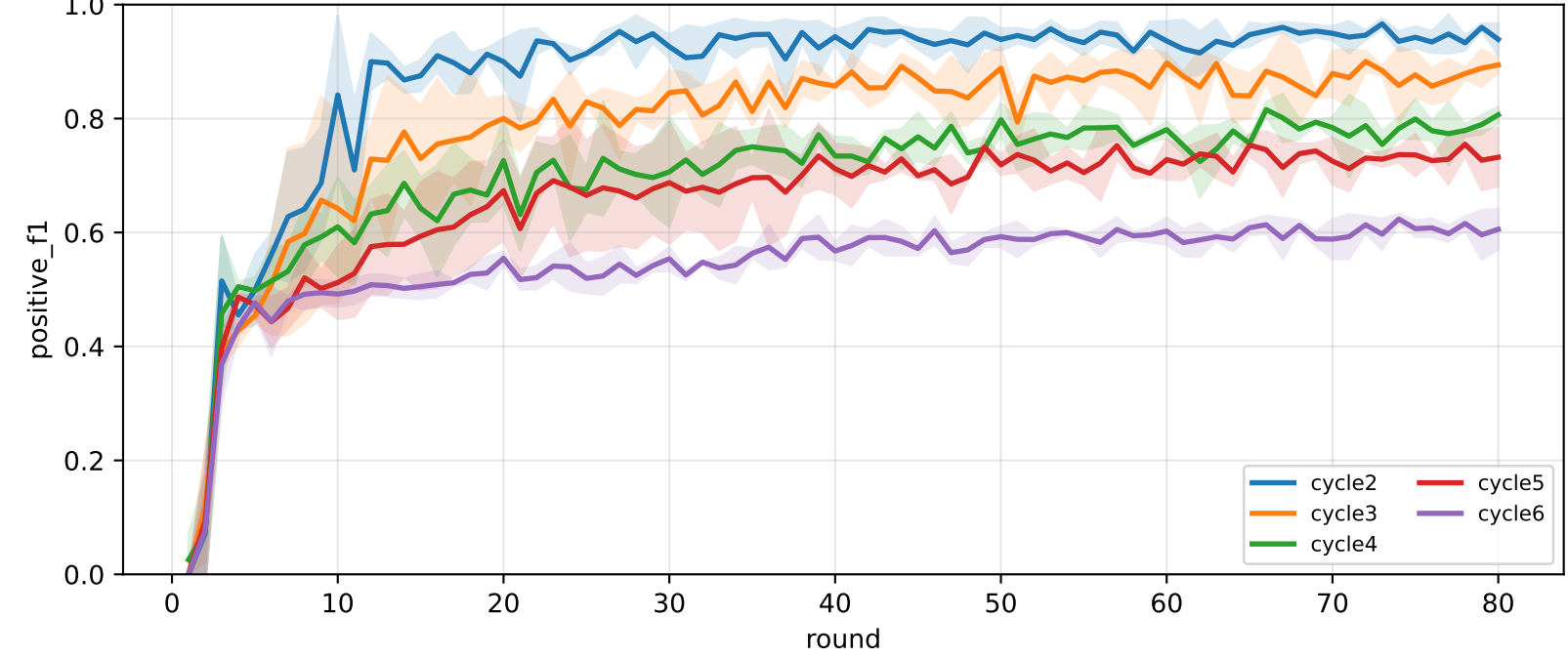
Pooled local centralized positive F1



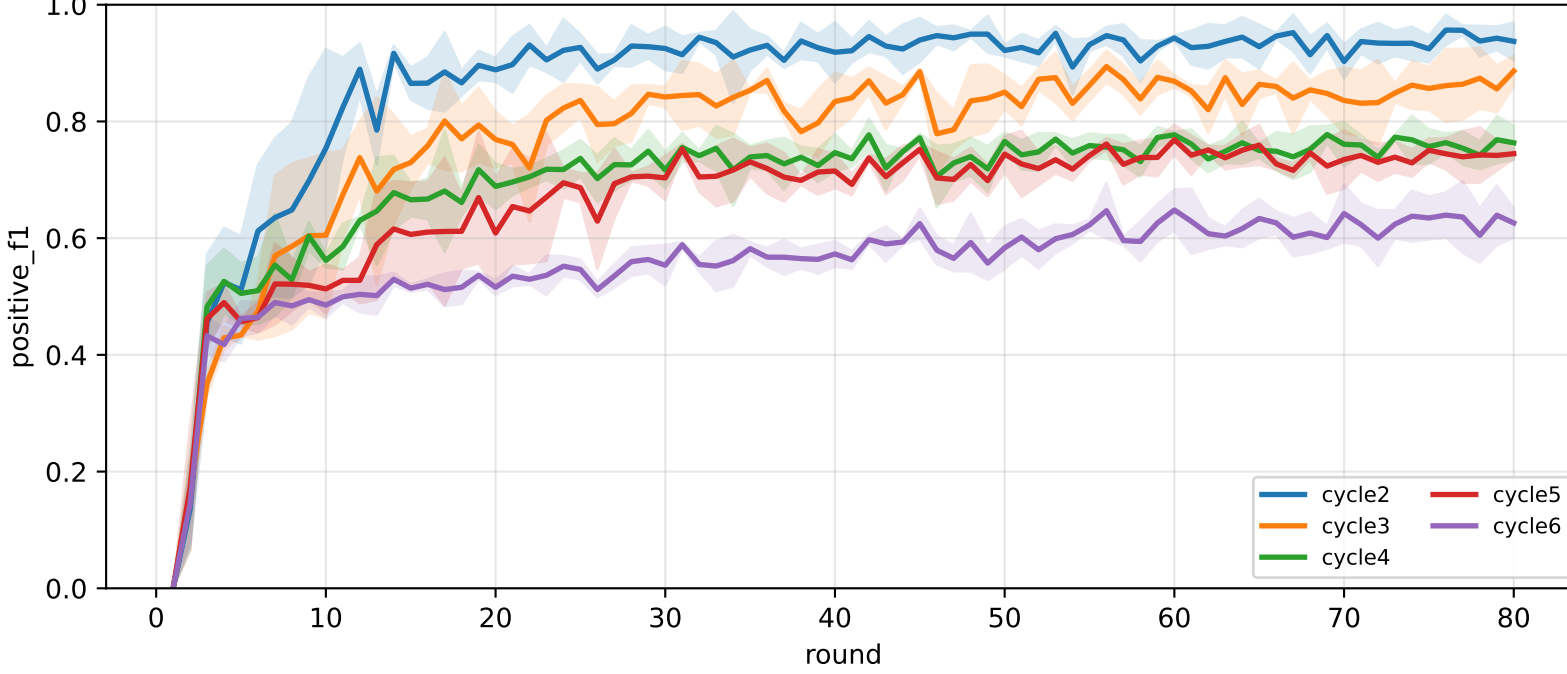
Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1

